

AI Restaurant Voice Assistant

An n8n backend that powers an ElevenLabs phone voice agent for a restaurant — checking table availability, booking, taking orders, and managing reservations, entirely by voice.



1 Executive Summary

Restaurant phone lines get flooded with the same handful of requests: “do you have a table for 4 at 8?”, “can I get a pickup order in?”, “I need to cancel my booking”. This backend gives an ElevenLabs voice agent five callable tools that handle exactly these requests, each backed by real capacity checking and validation so the agent never overbooks the dining room, orders never go unrecorded, and reservation changes always match an existing booking first.

2 Key Metrics

Trigger	5 webhooks called directly by ElevenLabs during a live phone call
Voice Platform	ElevenLabs Conversational AI — the agent calls these tools mid-conversation
Data Store	Google Sheets — Reservations, orders and Escalations tabs
Manager Notifications	Gmail, sent on every booking, order, cancellation, modification and escalation
Capacity Model	Fixed table capacity (40 covers) checked per date/time slot before confirming
Reservation Lookup	Cancel/modify requests are matched by phone number + date before any change
Pricing Safety	Orders are logged as received — total and timing are confirmed by a human, not the AI

5

VOICE TOOLS

40

COVERS CAPACITY

0

OVERBOOKED SLOTS

3 Business Value

- ✓ **Phone lines never busy** — reservations, orders and availability checks are handled by voice at any hour.
- ✓ **No overbooked dining room** — every reservation is checked against real capacity for that exact date and time first.
- ✓ **Orders never lost in translation** — pickup and delivery orders are logged verbatim and emailed to the kitchen immediately.
- ✓ **Manager always informed** — every booking, order, cancellation, modification and callback lands in their inbox in real time.

4 Architecture Overview

The voice agent is configured with five Webhook Tools, each pointing at its own n8n webhook.

TOOL	CALLED WHEN	PURPOSE
check_availability	Caller asks for a table	Sums booked party sizes for the slot and compares against total table capacity
make_reservation	Caller confirms a booking	Saves a confirmed booking to Reservations and emails the manager
place_order	Caller wants pickup/delivery	Saves the order to the orders tab and emails the manager
manage_reservation	Caller wants to cancel/modify	Finds a reservation by phone + date, then cancels or modifies it
request_human_callback	Anything out of scope	Logs an escalation and sends an urgent alert to the manager

5 How It Works

Check Availability

- 1 The requested date, time and party size are taken from the caller's request.
- 2 Every non-cancelled reservation for that exact slot is totalled by party size.
- 3 The request is compared against a fixed capacity of 40 covers.
- 4 The agent gets a clear yes/no, or a suggestion to try an earlier or later time.

Make a Reservation

- 1 Customer name, phone, party size, date, time and special requests are collected.
- 2 The booking is appended to the Reservations sheet.
- 3 A formatted email goes to the manager with the full booking detail.
- 4 The agent reads back a confirmation naming guest, party size, date and time.

Place an Order

- 1 Customer name, phone, items, order type and delivery address are collected.
- 2 The order is appended to the orders sheet.
- 3 The manager/kitchen receives a formatted email with the full item list.
- 4 The caller is told a team member will confirm total and timing shortly.

Manage Reservation

- 1 A non-cancelled booking is searched by phone number and date together.
- 2 If nothing matches, the caller is asked to double-check their details.
- 3 Cancel deletes the row; modify merges only the changed fields.
- 4 The agent receives a status of cancelled, modified, or not_found.

Human Callback

- 1 Customer name, phone number, reason and the conversation ID are recorded.
- 2 The request is appended to the Escalations tab.
- 3 The manager receives an urgent-styled email with a one-tap call link.

6 Reliability & Security Design

- ◆ **Real capacity checking** — availability is calculated from actual booked party sizes, not left to the AI to estimate.
- ◆ **Phone + date matching for changes** — cancelling or modifying requires both to match an existing booking.
- ◆ **Graceful “not found” handling** — a failed lookup asks the caller to double-check rather than guessing.
- ◆ **Pricing kept out of the AI's hands** — orders are logged as received, with total and timing confirmed by a human.
- ◆ **Full audit trail** — every reservation, order, cancellation, modification and escalation is logged and emailed.
- ◆ **Resilient notifications** — manager emails continue the workflow even if the send itself fails.

7 Setup Checklist

- ☒ Connect a Google Sheets credential with Reservations, orders and Escalations tabs.
- ☒ Connect a Gmail credential and replace the placeholder manager address on the five “Send” nodes.
- ☒ Set the restaurant's real table capacity in the Check Capacity node (default 40 covers).
- ☒ Activate the workflow and copy each webhook's Production URL into the matching ElevenLabs tool.
- ☒ Test all five tools end-to-end with a live or simulated call.

8 Node-by-Node Map

check_availability	Availability Webhook → Read Reservations → Check Capacity → Respond Availability
make_reservation	Reservation Webhook → Normalize Data → Log Booking → Send Manager Email → Respond
place_order	Order Webhook → Normalize Order → Log Order → Send Manager Email → Respond
manage_reservation	Manage Webhook → Find Matching Reservation → Cancel/Modify → Respond
request_human_callback	Escalation Webhook → Log Escalation → Send Manager Email → Respond

SUMMARY

Designed and built end-to-end in n8n, demonstrating a voice-agent backend for a restaurant with real-time capacity checking, order logging, and safe reservation management.